№ 2.18 с-1

$$2x_1^2+2x_1+4x_2-3x_3 \to extr$$

$$\begin{cases} 8x_1-3x_2+3x_3 \le 40 \\ -2x_1+x_2-x_3=-3 \\ x_2 \geq 0 \end{cases} \Rightarrow \begin{cases} 8x_1-3x_2+3x_2-40 \le 0 \\ -2x_1+x_2-x_3+3=0 \\ -x_2 \le 0 \end{cases} \Rightarrow$$

$$\begin{cases} 8x_1-3x_2+3x_3-40+v_1^2=0 \\ -2x_1+x_2-x_3+3=0 \\ -x_2+v_2^2=0 \end{cases}$$

$$\mathcal{L}(\bar{x},\bar{\lambda},\bar{v})=\lambda_0(2x_1^2+2x_1+4x_2-3x_3)+$$
$$+\lambda_1(8x_1-3x_2+3x_3-40+v_1^2)+\lambda_2(-x_2+v_2^2)+$$
$$+\lambda_3(-2x_1+x_2-x_3+3) \longrightarrow extr$$

$$\begin{cases} \frac{\partial \mathcal{L}}{\partial x_1}=4x_1+2+8\lambda_1-2\lambda_3=0 \\ \frac{\partial \mathcal{L}}{\partial x_2}=4-3\lambda_1-\lambda_2+\lambda_3=0 \\ \frac{\partial \mathcal{L}}{\partial x_3}=-3+3\lambda_1-\lambda_3 \\ \frac{\partial \mathcal{L}}{\partial \lambda_1}=8x_1-3x_2+3x_2-40+v_1^2=0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_2}=-x_2^2+v_2^2=0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_3}=-2x_1+x_2-x_3+3=0 \\ \frac{\partial \mathcal{L}}{\partial u_1}=2\lambda_1 v_1; \ \frac{\partial \mathcal{L}}{\partial u_2}=2\lambda_2 u_2; \ \lambda_0=1 \end{cases}$$

№ 2.18.с-2

①

$$\begin{cases} 4x_1 + 2 + 8\cdot 0 - 2\lambda_3 = 0 \\ 4 - 3\cdot 0 - \lambda_2 + \lambda_3 = 0 \\ -3 + 3\cdot 0 - \lambda_3 = 0 \\ 8x_1 - 3x_2 + 3x_3 - 40 + v_1^2 = 0 \\ -x_2 + v_2^2 = 0 \\ -2x_1 + x_2 - x_3 + 3 = 0 \\ 2\lambda_1 v_1 = 0 \\ 2\lambda_2 u_2 = 0 \\ \lambda_0 = 1;\ \lambda_1 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2 \\ \lambda_2 = 1 \\ \lambda_3 = -3 \\ x_2 = 0 \\ x_3 = 7 \\ v_1 = 0 \\ v = 0 \\ \lambda_0 = 1;\ \lambda_1 = 0 \end{cases} \quad (-2; 0; 7)$$

②

$$\begin{cases} 4x_1 + 2 + 8\lambda_1 - 2\lambda_3 = 0 \\ 4 - 3\lambda_1 - \lambda_2 + \lambda_3 = 0 \\ -3 + 3\lambda_1 - \lambda_3 = 0 \\ 8x_1 - 3x_2 + 3x_3 - 40 + 0 = 0 \\ -x_2 + v_2^2 = 0 \\ -2x_1 + x_2 - x_3 + 3 = 0 \\ 2\lambda_1 \cdot 0 = 0 \\ 2\lambda_2 u_2 = 0 \\ \lambda_0 = 1 \\ u_1 = 0 \end{cases} \Rightarrow \begin{cases} x_1 = -2 \\ \lambda_2 = 1 \\ \lambda_3 = -3 \\ x_2 = 0 \\ x_3 = 7 \\ \lambda_1 = 0 \\ u_2 = 0 \\ \lambda_0 = 1 \\ u_1 = 0 \end{cases}$$

№2.18 С-3

③

$$\begin{cases} 4x_1+2+8\lambda_1-2\lambda_3=0 \\ 4-3\lambda_1-\lambda_2+\lambda_3=0 \\ -3+3\lambda_1-\lambda_3=0 \\ 8x_1-3x_2+3x_3-40+u_1^2=0 \\ -x_2+u_2^2=0 \\ -2x_1+x_2-x_3+3=0 \\ 2\cdot 0=0 \\ 2\lambda_2 u_2=0 \\ \lambda_0=1 \\ \lambda_1, u_1=0 \end{cases} \Rightarrow \begin{cases} x_1=-2 \\ \lambda_2=1 \\ \lambda_3=-3 \\ x_2=0 \\ x_3=7 \\ \lambda_1=0,\ u_1=0 \\ u_2=0 \\ \lambda_0=1 \\ \lambda_1, u_1=0 \end{cases}$$

④

$$\begin{cases} 4x_1+2+8\lambda_1-2\lambda_3=0 \\ \left[\begin{array}{l} 4-3\lambda_1-0+\lambda_3=0 \\ -3+3\lambda_1-\lambda_3=0 \end{array}\right. \quad \varnothing \\ 8x_1-3x_2+3x_3-40+u_1^2=0 \\ -2x_1+x_2-x_3+3=0 \\ -x_2+u_2^2=0 \\ 2\lambda_1 u_1=0 \\ 2\cdot 0\cdot u_2=0 \\ \lambda_0=1 \\ \lambda_2=0 \end{cases}$$

$8x_1-3(x_2-x_3)-40$

$x_2-x_3=2x_1-3$

$x_2=0$

$\varnothing$

Нет решений

$u_2=0$

$\lambda_0=1$

$\lambda_2=0$

№2.18
С-4

(5)

$$\begin{cases} 4x_1+2+8\lambda_1-2\lambda_3 \\ \left[\begin{array}{l} 4-3\lambda_1-\lambda_2+\lambda_3 \\ -3+3\lambda_1-\lambda_3=0 \end{array}\right. \quad \varnothing \\ 8x_1-3x_2+3x_3-40+u_1^2=0 \\ -2x_1+x_2-x_3+3=0 \\ -x_2+u_2^2=0 \\ 2\lambda_1 u_1=0 \\ 2\lambda_1\cdot 0=0 \\ \lambda_0=1 \\ u_2=0 \end{cases}$$

$\varnothing$

Нет решений

$-2x_1-x_3+3=0$

$x_2=0$

$\lambda_2=0$

$\lambda_0=1$

$u_2=0$

(6)

$$\begin{cases} 4x_1+2+8\lambda_1-2\lambda_0=0 \\ 4-3\lambda_1-\lambda_2+0=0 \\ -3+3\lambda_1-0=0 \\ 8x_1-3x_2+3x_3-40+u_1^2=0 \\ -2x_1+x_2-x_3+3=0 \\ -x_2+u_2^2=0 \\ 2\lambda_1 u_1^2=0 \\ 2\lambda_2 u_2=0 \\ \lambda_0=1 \\ \lambda_3=0 \end{cases}$$

$x_1=-\frac{5}{2}$

$\lambda_2=1$

$\lambda_1=1$

$x_3=8$

$x_2=0$

$\left(-\frac{5}{2};0;8\right)$

$u_1=0$

$u_2=0$

$\lambda_0=1$

$\lambda_3=0$

7

№2.18
С-5

$$\begin{cases} 4x_1 + 2 + 8\lambda_1 - 2\lambda_3 = 0 \\ 4 - 3\lambda_1 - 0 + \lambda_3 = 0 \\ -3 + 3\lambda_1 - \lambda_3 = 0 \\ 8x_1 - 3x_2 + 3x_3 - 40 - u_1^2 = 0 \\ -x_2 + 0 = 0 \\ 2\lambda_1 u_1 = 0 \\ 2 \cdot 0 \cdot 0 = 0 \\ \lambda_0 = 1 \\ \lambda_2, u_2 = 0 \end{cases}$$

$\varnothing$

Нет решений

Ответ: $(-2; 0; 7)$, $\left(-\frac{5}{2}; 0; 8\right)$

$2 \cdot (-2)^2 + 2 \cdot (-2) + 4 \cdot 0 - 3 \cdot 7 = 2 \cdot 4 - 4 + 0 - 21 =$

$= 8 - 4 - 21 = -17$